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CHAPTER NINETEEN

Audition

Henry E. Heffner and Rickye S. Heffner

Introduction

Fifty years ago, a general review of auditory research could be accomplished in four chapters (Stevens, 1951); today, one review has grown to 14 volumes with more on the way (*Springer Handbook of Auditory Research*, R. R. Fay and A. N. Popper, series editors). The growth of auditory research has come about because more areas of neuroscience have been applying their techniques to understanding the neurological basis of hearing. Although a complete understanding requires a multidisciplinary approach, it is often the case that researchers know little about important issues in closely related disciplines – a problem that is becoming widespread in neuroscience (Cahill, McGaugh, & Weinberger, 2001). This problem has been compounded by an emphasis on recent research, giving those new to the field the impression that there is little to be learned from older work (e.g., Moore, Rothholtz, & King, 2001).

One area of auditory research that impinges on all others is the behavioral study of hearing in animals. Because all anatomical and physiological models of auditory processing must eventually be related to behavior, and because virtually all such models are based on animal research, it is obviously necessary to know the hearing abilities of animals. Furthermore, it should be possible to test the validity of such models by studying the effects of central nervous system lesions on hearing, that is, ablation/behavior experiments.

The purpose of this chapter is to describe two lines of behavioral research. The first is the comparative study of mammalian hearing, that is, the determination of *what* mammals hear and *why* they hear as they do. The second is the study of the auditory cortex using the ablation/behavior approach. In doing so, we will highlight issues to be considered when interpreting and applying this research.

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Comparative Study of Mammalian Hearing

The first comparative studies of hearing were conducted in the nineteenth century by Francis Galton, who used specially constructed whistles to determine the unconditioned responses of animals to high-frequency sounds (Galton, 1883). Galton made several discoveries that have since been supported by modern research, such as that cats have particularly good high frequency hearing and that human high-frequency hearing ability declines with age. However, he incorrectly concluded that large dogs could not hear high frequencies, because, unlike small dogs, they showed no reaction to his whistles (H. E. Heffner, 1983). Thus he was unable to distinguish the inability to hear a sound from the failure to respond to it, although he was well aware of such a possibility. This is the problem with using a simple startle reaction as a test of hearing.

Modern comparative studies of mammalian hearing have focused on the basic auditory abilities of detection, localization, and frequency discrimination. (Interest in the ability to perform more complex auditory discriminations is growing, e.g., Dooling & Hulse, 1989.) Before turning to what we know about these basic abilities, it is important to consider how they are measured.

Behavioral measurement of hearing

A number of behavioral procedures for determining the sensory abilities of animals were available by the late 1960s (e.g., Stebbins, 1970). These procedures, along with the use of precision instruments for presenting and measuring sound, made it possible to use conditioning techniques to determine the behavioral hearing abilities not only of mammals, but birds and fish as well (Klump, Dooling, Fay, & Stebbins, 1995). Reptiles, on the other hand, have proven virtually impossible to condition to sound (for an exception, see Patterson, 1966), although they readily learn to make visual discriminations (e.g., Burghardt, 1977). Similarly, no conditioning procedures have been developed for amphibians, although the use of startle reflex modification techniques and the natural tendency of females during mating season to approach the sound of a male have been used to study some aspects of frog hearing (e.g., Klump et al., 1995). The fact that amphibians and reptiles do not readily learn to respond to sound suggests that they may lack the neurological mechanisms for doing so.

Some conditioning procedures are better than others. The better ones not only reward an animal for correctly responding to a sound, but they also have good control over responses in the absence of that sound. In the language of signal detection theory, this fact means that "hits" are sufficiently rewarded and "false positives" are kept under control. A good procedure should also be easy for an animal to learn and should allow their heads to be fixed in the sound field so that the stimulus reaching their ears can be specified with some precision. One procedure that meets these requirements is "conditioned suppression," in which an animal is trained to place its mouth on a spout in order to receive water (or food), and to break contact with that spout whenever a stimulus is presented that signals impending shock (H. E. Heffner & Heffner, 1995a).

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By carefully adjusting the reward rate and shock intensity, it is possible to optimize an animal's performance – that is, to maximize its hit rate and minimize its false positive rate. Animals are capable of learning the basic avoidance response within a session or two and the response of placing its mouth on a spout fixes the animal's head within the sound field. Although procedures that use positive reward with a delay or "error time out" as a punisher may work well in some cases, comparisons have shown that shock is a more effective punisher and the combined use of positive reward and shock generally gives better results (e.g., H. E. Heffner & Heffner, 1984).

In a hearing test, sounds are presented to an animal either from a loudspeaker or through headphones. Although headphones make it possible to test each ear separately, problems may arise in calibrating the sound (e.g., Pfingst, Hienz, & Miller, 1975; Zhou & Green, 1995). Moreover, headphones bypass the external ear and thus will not reflect the contribution of the pinnae to hearing. Thus, for comparative studies, sounds are best presented from a loudspeaker located in front of the animal in an acoustic environment that minimizes sound reflections, that is, a free field (Larsen, 1995).

In measuring an animal's performance, it is necessary to correct its hit rate for false positives (e.g., H. E. Heffner & Heffner, 1985, 1995a). One way to do this is to reduce the hit rate in proportion to the false positive rate using the formula: "Corrected Hit rate = Hit rate − (Hit rate × False Positive rate);" this formula is sometimes expressed as "Hit Rate × (1 − False Positive rate)." Another way to correct for false positives is to calculate an animal's percentage correct using the formula: "(Hit rate + (1 − False Positive rate))/2"; note that 1 − False Positive rate is known as the "correct rejection rate." The threshold for a particular discrimination is then defined as the stimulus that gives a corrected hit rate of 0.50 or a percentage correct of 75 percent. Experience has shown that thresholds defined in this way remain stable over a range of false positive rates.

Although it has been claimed that performance should be specified using a signal detection measure such as d' or A' (e.g., Penner, 1995), there are at least two reasons for avoiding such measures when working with animals. First, the values generated by these calculations are nonintuitive and cannot be interpreted without additional information. For example, a corrected hit rate of 50 percent means that the animal is capable of detecting a signal half of the time and conveys more information than the statement that its d' is 2.33 or 1.64 (which are the d' values for a 50% hit rate with 1% and 5% false alarm rate, respectively). Second, it is sometimes assumed that a researcher can use such measures to obtain useful information when an animal has a high false positive rate. However, a high false positive rate (e.g., > 25%) may indicate that the animal is not carefully attending to the stimulus and is attempting to perform the task by guessing, a situation in which signal detection measures do poorly (Green, 1995).

Detection of sound

The most basic measure of hearing is an animal's *behaviorally determined* sensitivity to pure tones throughout its hearing range, that is, its audiogram. Although electrophysiological estimates of absolute sensitivity may be of interest in their own right, they do not accurately reflect behavioral sensitivity. This applies to such electrophysiological measures

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Figure 19.1 Human audiogram illustrating the 60-dB high- and low-frequency hearing limits, best frequency, and best sensitivity.
Source: Adapted from H. E. Heffner & R. S. Heffner (1992), with permission.

as the cochlear microphonic, thresholds of inferior colliculus neurons, and the auditory brainstem response (see below).

The audiograms of animals are compared on the basis of the following features: high- and low-frequency hearing limits, frequency of best hearing, and best sensitivity (see Figure 19.1). Of these measures, the high- and low-frequency hearing limits have proven to be the most interesting. Note that these limits are defined as the highest and lowest frequencies audible at a particular intensity level.

High-frequency hearing Mammals differ from other vertebrates in that virtually all of them hear well above 10 kHz, which is the upper limit for birds. Reptiles, amphibians, and fish (with some exceptions) do not hear above 5 kHz (H. E. Heffner & Heffner, 1998). Those few mammals that do not hear above 10 kHz (i.e., subterranean mammals) are considered to have degenerate hearing (R. S. Heffner & Heffner, 1993).

High-frequency hearing ability varies between different species of mammals – the 60-dB high-frequency hearing limit ranges from 5.9 kHz for the blind mole-rat (a subterranean rodent) to over 100 kHz for some bats and porpoises, a range of more than 4.5 octaves (H. E. Heffner & Heffner, 1998). There is substantial evidence that high-frequency hearing evolved in mammals primarily for the purpose of localizing sound; although some species also use high-frequency communication calls, and bats also use their high-frequency hearing for echolocation, these uses appear to be secondary adaptations.

The existence of variation in the high-frequency hearing of mammals came to the attention of the late R. Bruce Masterton in 1967 when he noticed that smaller mammals generally had better high-frequency hearing than larger ones. Because he was studying

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sound localization at the time, he realized that this observation had implications for the use of the two binaural sound-localization cues: the difference in the time of arrival of a sound at the two ears (Δt) and the difference in the frequency-intensity spectra of the sound reaching the two ears (Δfi). Noting that the magnitude of the Δt cue decreases with the size of an animal's head, he suggested that the smaller an animal's head, the more dependent it would be on the Δfi cue (Masterton, Heffner, & Ravizza, 1969). However, for an animal to use the Δfi cue, it would need to hear frequencies high enough to be effectively shadowed by its head and pinnae because small heads do not block lower frequencies as effectively as large heads. Therefore, the smaller an animal's *maximum* Δt (the time it takes for sound to travel around the head from one ear to the other), the higher it must hear to adequately localize sound.

The relation between maximum Δt (sometimes called functional head size) and high frequency hearing has proven to be robust ($r = -0.79$, $p < 0.001$) and has been shown to hold for over 60 animals (R. S. Heffner, Koay, & Heffner, 2001a). The only modification to this theory has been the realization of the importance of high-frequency hearing for pinna cues as well as for Δfi. Over the years, the work of Bob Butler and others has demonstrated that the directionality of the pinna not only provides effective cues for localizing sound in the horizontal plane, but it also provides the primary cues for vertical localization and for preventing front–back confusions (e.g., Butler, 1975; Musicant & Butler, 1984). For pinna cues to be effective in humans, the sounds must contain frequencies above 4 kHz and even sounds as high as 15 kHz have been shown to be necessary for optimal localization performance. Thus, the upper two octaves of human hearing (from 4 to 16 kHz) appear to be used primarily, if not exclusively, for sound localization as they are not necessary for the perception of speech. As with the Δfi cue, the smaller the pinnae, the higher an animal must hear in order to extract pinnae cues for sound localization.

The importance of high-frequency hearing for sound localization is also supported by two lines of experimental evidence. First, filtering out high frequencies from a signal has been shown to degrade sound localization performance in monkeys, humans, horses, chinchillas, and mice (Brown, 1994; Butler, 1975; H. E. Heffner & Heffner, 1983; R. S. Heffner, Heffner, & Koay, 1995; R. S. Heffner, Koay, & Heffner, 2001b). Thus mammals require high-frequency hearing to localize sound using either the Δfi locus cue, pinna cues, or both. Second, it appears that subterranean animals that are adapted to the one-dimensional world of an underground habitat have little use for sound localization and are therefore released from the selective pressure to hear high frequencies. Thus, the pocket gopher, naked mole-rat, and blind mole-rat do not localize sound and have lost their high-frequency hearing as well as their pinnae (R. S. Heffner & Heffner, 1990, 1992b, 1993). In short, sound localization and high-frequency hearing go hand-in-hand in mammals. Mammals cannot adequately localize sound without high frequencies and those that relinquish the ability to localize sound also give up their high frequency hearing.

Low-frequency hearing The variation in mammalian low-frequency hearing is even greater than that for high-frequency hearing. Indeed, the 60 dB low-frequency limit for mammals extends from 17 Hz (the Indian elephant) to 10.3 kHz (the little brown bat), a

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range of over nine octaves. Analysis of this variation has resulted in two findings. First, mammals appear to fall into two groups: those that hear below 125 Hz and those that do not. Second, low-frequency hearing varies with high-frequency hearing (R. S. Heffner et al., 2001a).

Figure 19.2 shows the distribution of 60 dB low-frequency hearing limits for mammals (underwater audiograms have been excluded because of the difficulty in equating air and water thresholds). Of the 59 species, 38 have low-frequency hearing limits below 125 Hz whereas 20 species have low-frequency hearing limits above 500 Hz. Only one species falls within the two-octave gap from 125 to 500 Hz, the subterranean pocket gopher (an animal with degenerate hearing).

We have suggested that the two groups may differ in the mechanisms they use to perceive the pitch of a sound (R. S. Heffner et al., 2001a). Briefly, there are two different neural mechanisms that may underlie the perception of pitch (for a recent description, see Moore, 1993). In one mechanism, frequency is encoded by temporal mechanisms based on phase locking of auditory nerve fibers; this mechanism is limited to low frequencies because phase locking declines as frequency increases. In the second mechanism, higher frequencies are encoded by a place mechanism in which tones of different frequencies excite hair cells and nerve fibers at different locations along the basilar membrane. However, the actual frequencies over which either the temporal or the place mechanism is dominant are not agreed upon. Some observations suggest that the upper limit of the temporal mechanism for the perception of pitch is around 4–5 kHz (e.g., Moore, 1993). However, other observations, such as studies of the perception of the pitch of click trains and psychophysical studies of patients with cochlear implants, suggests that temporal coding extends up to only about 300 Hz (Flanagan & Guttman, 1960; Shannon, 1983). Because this latter upper limit corresponds to the 125–500 Hz gap in mammalian low-frequency limits, it suggests that the animals that do not hear below 500 Hz are not using temporal coding for pitch perception. Thus, animals that hear below 125 Hz may be using both temporal and place mechanisms, whereas those that do not hear below 500 Hz may be using only the place mechanism.

Various factors, such as body size, phyletic lineage, and lifestyle have been examined in an attempt to explain the variation in low-frequency hearing. So far, the only factor found to be reliably correlated with low-frequency hearing is high-frequency hearing – that is, animals with good high-frequency hearing generally have poor low-frequency hearing (R. S. Heffner et al., 2001a). The degree to which high- and low-frequency hearing are related differs for the two groups of mammals: among those that do not hear below 500 Hz, $r = 0.691$ ($p = 0.0015$), whereas for those that hear below 125 Hz, $r = 0.567$ ($p = 0.0006$). In either case, high-frequency hearing accounts for considerable variance in low frequency hearing for both groups, suggesting that it may provide a clue for understanding some of the variation in low-frequency hearing.

The existence of such a relationship suggests that good high- and low-frequency hearing are incompatible. One possibility is that there is some anatomical limitation that prevents the mammalian ear from encoding both high and low frequencies. However, there are several species with good high- and low-frequency hearing, including the chipmunk (39 Hz to 52 kHz), least weasel (50 Hz to 60 kHz), bushbaby (92 Hz to

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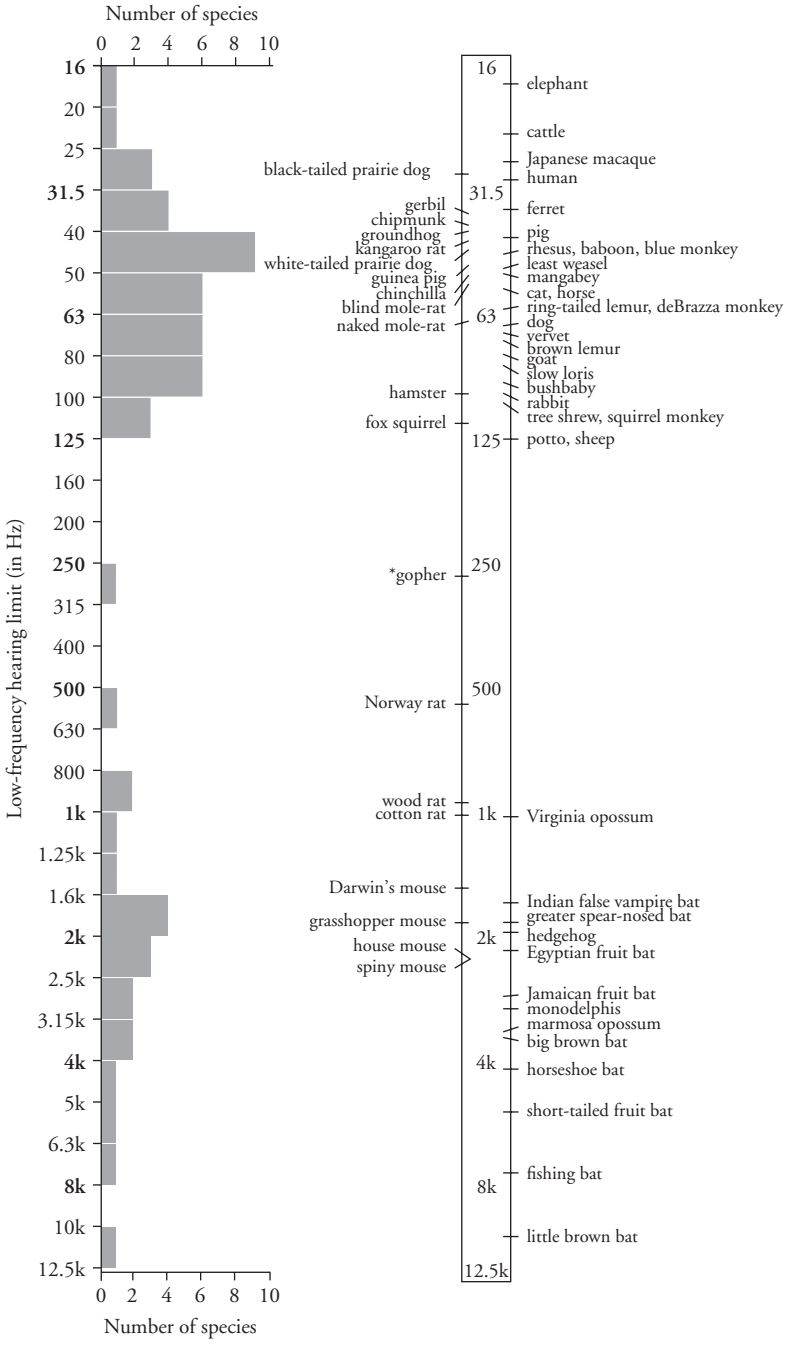


Figure 19.2 Distribution of mammalian low-frequency hearing limits (lowest frequency audible at 60 dB). Note that low-frequency hearing limits fall into two distinct groups with a gap between them at 125 to 500 Hz. Rodents (shown on the left side of the bar) are the only order with members in both groups.

* Note that only the subterranean gopher, which has vestigial hearing, falls into the gap between 125 and 500 Hz. Bin width of the histogram is 1/3 octave.

Source: Data from R. S. Heffner et al. (2001a).

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65 kHz), and domestic cat (55 Hz to 79 kHz), suggesting that any anatomical constraints that might exist can be overcome. Another possibility is that good low-frequency hearing could be disadvantageous in situations where low-frequency sounds interfere with the analysis of high-frequency sounds. For example, we have noted that animals often localize high-frequency noise slightly more accurately than broadband noise (R. S. Heffner et al., 1995). This finding suggests that mammals may restrict their low-frequency hearing to prevent the low-frequency component of sounds from interfering with the analysis of the high frequency components needed for sound localization.

Localization of sound

At one time it was believed that all mammals were under selective pressure to localize sound as accurately as possible, and that the only factor limiting their accuracy was the availability of the locus cues (as determined by head size). However, it is now clear that animals localize sound only as accurately as is necessary to direct their eyes to the source of a sound. Moreover, an animal may relinquish one or more of the sound localization cues, or even the entire ability to localize sound, if it is not needed for survival.

Measuring sound localization ability There are two ways to measure sound localization ability. One is to have the subject point in the direction from which a sound appeared to come and measure the *accuracy* of pointing. The other is to have the subject discriminate between two sound sources, bringing the sources closer together until they can no longer be discriminated, a procedure that measures localization *acuity*.

Accuracy of sound localization can be measured by training an animal to orient its head toward the source of a sound. This procedure is commonly used with owls as they have a strong natural orientation response to sound that can be maintained with food reward (e.g., Wagner, 1995). Among mammals, only cats have been successfully trained to orient to the source of a sound (e.g., Populin & Yin, 1998). Most mammals do not reliably point their head to a sound source and there are several reasons why this may be: first, their visual fields may be so broad that they do not need to turn their heads much to see the sound source; second, they may be able to turn their eyes instead of their head; and finally, their natural response to an unexpected sound may be to freeze.

One way to measure localization acuity is to train an animal to discriminate between two sound sources by walking to the source of a sound to receive a reward (e.g., Neff, Diamond, & Casseday, 1975; Thompson, Heffner, & Masterton, 1974). Another common method is to have the animal discriminate between two sound sources by responding when a sound comes from one location but not another (e.g., Brown, 1994; H. E. Heffner & Heffner, 1984). Using the conditioned suppression task, an animal can be trained to maintain contact with a water spout to receive water while a sound is presented from one location, and to break contact to avoid a mild electric shock when the sound comes from a different location. In either case, the angle of separation between the two sound sources is then reduced until the animal is no longer able to discriminate between the two sources.

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Both accuracy (pointing) and acuity (discrimination) measures give essentially the same results. However, fine-grain comparisons of the two procedures are complicated because accuracy is measured in terms of how much the subject's estimate of location differs from true location whereas acuity is measured in terms of "minimum audible angle," that is, the smallest angle that can be discriminated (for a recent comparison of the two procedures, see Recanzone, Makhamra, & Guard, 1998). However, minimum audible angle is a better comparative measure of sensory ability because it is not confounded by species differences in the ability to point.

The most common sound-localization measure is the minimum audible angle for a left–right discrimination, in which the animal is trained to discriminate two sound sources located in front of, and centered on, its midline. The standard stimulus is a 100 ms broadband noise burst, which is too brief to be tracked or scanned, but contains both high and low frequencies, thus permitting the use of all three sound-localization cues: binaural time difference (Δt), binaural frequency-intensity spectral difference (Δfi), and pinna cues.

In addition to determining acuity around midline, minimum audible angle can be determined for locations off to the side, including centering the sound sources on the interaural axis (i.e., front–back localization) and for vertically separated sound sources (elevation). Both the front–back and vertical-localization tests measure the ability of an animal to localize in situations where the pinnae provide the primary cues for discriminating locus.

In testing an animal's ability to discriminate two sound sources, it can be difficult to obtain loudspeakers that are perfectly matched for broadband noise. As a result, an animal may shift from discriminating locus to discriminating the quality differences in loudspeakers when the angle of separation is too small to distinguish locus. This problem can be avoided by using several pairs of loudspeakers during a session, by randomizing the intensity of the sound, and by not testing at subthreshold angles for extended periods. However, the crucial test of whether an animal is discriminating sounds on the basis of locus, as opposed to speaker quality, is to demonstrate that there is some small angle at which the animal performs at chance – it is not sufficient to assume that performance would fall to chance if both speakers were placed at 0°.

Variation in sound localization acuity When early sound-localization studies showed that humans and elephants have better sound-localization acuity than cats and dogs, which in turn have better acuity than rats, it was naturally assumed that the large binaural cues generated by large heads were necessary for good localization. This belief was abandoned when it was subsequently discovered that horses and cattle have poorer acuity than rats (R. S. Heffner & Heffner, 1992a).

Midline sound localization thresholds have been obtained for over 30 different species of mammals, from mice to elephants (H. E. Heffner & Heffner, 1998; R. S. Heffner, Koay, & Heffner, 2001b). Thresholds range from about 1° for humans and elephants to more than 20° for horses and cattle, and over 30° for house mice with the subterranean rodents (gopher, blind mole-rat, and naked mole-rat) being unable to localize brief sounds (see Table 19.1).

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Table 19.1 Sound localization acuity and use of binaural cues in mammals

Species	*Midline acuity*	*Binaural phase cue*	*Binaural intensity cue*
Indian elephant	1.2°	yes	yes*
Human	1.3°	yes	yes
Harbor seal	3.2°	—	—
Domestic pig	4.5°	yes	no
Virginia opossum	4.6°	—	—
Domestic cat	5.2°	yes	yes
Squirrel monkey	5.9°	—	—
Japanese macaque	6.8°	yes	yes
Dog	8.0°	—	—
Sea lion	8.8°**	—	—
Jamaican fruit bat	9.9°	yes	yes
Greater spear-nosed bat	10°	no	yes
Egyptian fruit bat	11.6°	yes	yes
Ferret	11.8°	—	—
Least weasel	12°	yes	yes
Wild Norway rat	12.2°	yes	yes
Fox squirrel	14°	yes	yes
Big brown bat	14°	no	yes
African pigmy hedgehog	14.3°	no	yes
Short-tailed fruit bat	14.5°	no	yes
Chinchilla	17.5°	yes	yes
Domestic goat	18°	yes	yes*
Hamster	18.8°	—	—
Spiny mouse	18.9°	no	yes
Desert hedgehog	19°	no	yes
Wood rat	19°	—	—
Grasshopper mouse	19.3°	no	yes
Domestic rabbit	22.3°	—	—
Horse	25°	yes	no
Cottontail rabbit	27°	—	—
Gerbil	27°	yes	yes
Merriam's kangaroo rat	27°**	yes	yes
Groundhog	27.8°	yes	yes
Cattle	30°	yes	no
Black-tailed Prairie dog	32.8°	yes	yes
Eastern chipmunk	33°	yes	yes
House mouse	33°	no	yes

Dashes indicate no data.

* Unable to use binaural intensity cue in the upper octaves of its hearing range.

** Localization tested with clicks. For all other animals, the stimulus was broadband noise.

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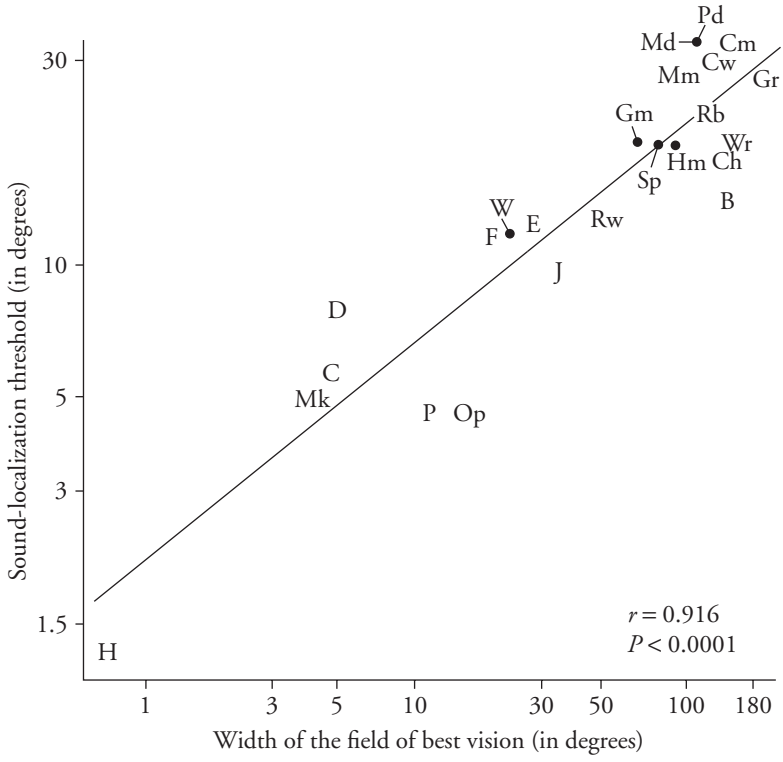


Figure 19.3 Relation between the width of the field of best vision and sound-localization threshold. The field of best vision is defined anatomically as the area of the retina containing ganglion-cell densities at least 75% of maximum. Species with narrow fields of best vision have better localization acuity (smaller thresholds) than species with broad fields of best vision. **B**, big brown bat; **C**, domestic cat; **Ch**, chinchilla; **Cm**, chipmunk; **Cw**, cow; **D**, dog; **E**, Egyptian fruit bat; **F**, ferret; **Gm**, grasshopper mouse; **Gr**, gerbil; **H**, human; **Hm**, hamster; **J**, Jamaican fruit bat; **Md**, domestic mouse; **Mk**, Japanese macaque; **Mm**, marmot; **Op**, Virginia opossum; **P**, domestic pig; **Pd**, prairie dog; **Rb**, domestic rabbit; **Rw**, wild normal rat; **Sp**, spiny mouse; **W**, least weasel; **Wr**, wood rat.
Source: Data from R. S. Heffner et al. (2001c).

The explanation for the variation in mammalian sound localization acuity lies in the fact that the primary function of sound localization is to direct the eyes to the source of a sound (R. S. Heffner & Heffner, 1992c). Just how accurate sound localization must be to direct the eyes depends on the width of an animal's field of best vision. Animals with narrow fields of best vision, such as humans, require good sound-localization acuity to direct their gaze so that the image of the sound source falls upon their field of best vision (e.g., the human fovea), whereas animals with broad fields, such as those with visual streaks, do not require as high a degree of sound-localization acuity to direct their gaze.

The relation between the width of the field of best vision (estimated from retinal ganglion cell densities) and sound-localization acuity is shown in Figure 19.3. This

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figure illustrates that mammals with narrow fields of best vision are more accurate localizers than mammals with broader fields ($r = 0.916$). The close relation between vision and sound localization is further supported by the observation that the subterranean rodents, which are adapted to living in dark burrows where visual scrutiny of sound sources is not possible, have lost virtually all of their ability to localize sound.

Use of binaural locus cues Mammals vary not only in sound localization acuity, but also in their use of binaural time (Δt) and binaural spectral (Δfi) locus cues. Whereas most mammals use both binaural cues, some use only one or the other. Animals that differ in their use of these cues should show corresponding differences in the physiology of their auditory systems.

The ability to use Δt and Δfi to localize sound can be determined two ways (e.g., R. S. Heffner & Heffner, 1992a). The first is to use headphones to present sounds separately to each ear and varying the relative time of arrival or intensity of the sound at the two ears. However, this test can only be used with animals that can be fitted with headphones. The second method is to present pure tones from two loudspeakers located in front of the animal at a fixed angle of separation and determine its ability to localize low- and high-frequency pure tones. This test can be used with any animal and is based on the fact that low-frequency pure tones are localized using binaural time-difference cues whereas high frequencies are localized using binaural intensity-difference cues. Briefly, low-frequency pure tones that bend around the head with little or no attenuation can only be localized by comparing the time of arrival of the phase of each cycle of the tone at the two ears, the binaural phase difference cue being a subset of Δt. However, the phase-difference cue becomes ambiguous for pure tones at high frequencies when successive cycles arrive too quickly for the nervous system to match the arrival of the same cycle at the two ears. The exact "frequency of ambiguity" depends on an animal's head size and the angle of the sound source relative to its midline – it is higher for smaller heads and sound sources closer midline (e.g., R. S. Heffner et al., 2001c). Pure tones above the frequency of ambiguity, then, must be localized using the binaural intensity-difference cue, a subset of Δfi. Thus, the ability of an animal to use the two binaural cues can be measured by determining the ability to localize pure tones above and below the frequency of ambiguity.

Most studies of the use of binaural cues have determined an animal's ability to localize pure tones presented from loudspeakers placed 30° to the left and right of its midline. The results of these studies have shown that the majority of mammals are able to localize both low- and high-frequency pure tones, indicating that they can use both binaural phase- and binaural intensity-difference cues. However, some animals can use only one of these cues (see Table 19.1). For example, horses and cattle can use binaural time, but not binaural intensity, whereas house mice and big brown bats use binaural intensity, but not binaural time. Finally, a few animals, such as the goat and Indian elephant, use both cues, but are unable to use binaural intensity differences for frequencies in the upper octaves of their hearing ranges.

Another aspect of sound localization for which mammals show systematic differences is in the upper frequency limit for use of the binaural phase cue. As shown in Figure 19.4,

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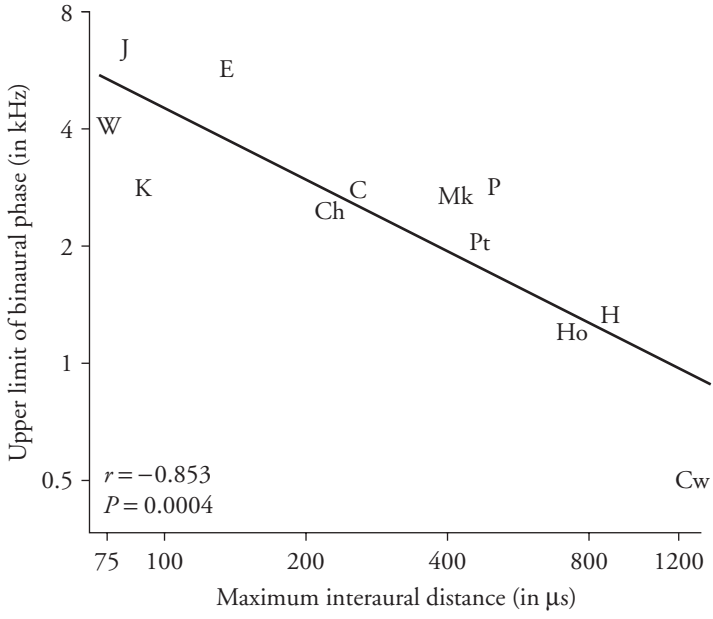


Figure 19.4 Relation between maximum interaural distance and the highest frequency at which the use of the binaural phase-difference cue has been observed. Although the binaural phase-difference cue is physically available at higher frequencies for animals with smaller heads, the limits shown here represent an animal's behavioral upper limit, not physical availability. **C**, domestic cat; **Ch**, chinchilla; **Cw**, cow; **E**, Egyptian fruit bat; **H**, human; **HO**, horse; **J**, Jamaican fruit bat; **K**, kangaroo rat; **Mk**, Japanese macaque; **P**, domestic pig; **Pt**, pig-tailed macaque; **W**, least weasel.
Source: Data from R. S. Heffner, et al. (2001c).

the upper frequency limit for the use of binaural phase spans a range of more than three octaves from the 500 Hz upper limit of cattle to the 6.3 kHz upper limit of the Jamaican fruit bat. Specifically, animals with small heads (small interaural distances) are able to use the binaural phase cue at higher frequencies than animals with larger heads. Although this relation seems obvious because the phase-difference cue is physically available at higher frequencies for animals with smaller heads, it should be noted that the upper limits shown in Figure 19.4 are, in most cases, well below the frequency of ambiguity. Thus they represent the ability of the animals' auditory systems to extract the binaural phase cue. Because using the binaural phase cue requires that auditory neurons fire in synchrony with the phase of the sine wave (i.e., phase lock), the variation in the upper limit for using binaural phase suggests that there is also variation in the upper limit of phase locking. Interestingly, the Jamaican fruit bat appears capable of phase locking up to at least 6.3 kHz, which is higher than the 5 kHz commonly listed as the upper limit for phase locking in the mammalian auditory system (R. S. Heffner et al., 2001c; Moore, 1997).

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Discrimination of frequency

The ability to discriminate frequency has been determined for a small number of species (Fay, 1988). The most common procedure for obtaining frequency discrimination thresholds is to train an animal to discriminate between a standard tone and a comparison tone of higher frequency and then reducing the frequency of the comparison tone until the animal can no longer discriminate between the two. Frequency discrimination thresholds are then obtained at frequencies throughout the animal's hearing range.

So far, comparison of the abilities of different species to discriminate frequency has not yielded any theoretical insight (e.g., Fay, 1992). A possible explanation for this result may be that the way in which frequency-difference thresholds are obtained does not always yield an accurate estimate of an animal's ability. Some time ago we noticed that some animals appeared to have unusual difficulty performing frequency discriminations, as compared to their performance on detection and localization tasks. Indeed, even an animal as intelligent and as cooperative as the Indian elephant showed poorer asymptotic performance when discriminating frequency (R. S. Heffner & Heffner, 1982). It is possible that the difficulty some animals have in performing a frequency discrimination is due to the way in which the stimuli are presented. That is, it is often easier to train an animal to detect when a tone is changing in frequency than it is to train it to discriminate between discrete tones that differ in frequency. Moreover, there are a number of natural sounds in which the direction of a frequency change is a significant parameter (e.g., communication calls), suggesting that animals may naturally be more experienced in detecting such changes. Thus, a more appropriate test of the ability to discriminate frequency may be to determine an animal's ability to detect frequency changes, such as frequency sweeps and frequency-modulated tones.

Final considerations regarding mammalian hearing

It is necessary to emphasize both the importance of knowing what animals hear and that such information must be obtained behaviorally.

General relevance of species differences in hearing Given the differences in mammalian hearing ranges, sounds that are clearly audible to one species may be completely inaudible to another. An extreme example is the Indian elephant, which hears up to 11.8 kHz, and the little brown bat, which hears down to 10.3 kHz – their hearing ranges show such little overlap that they hear virtually no sounds in common (H. E. Heffner & Heffner, 1998). However, even common laboratory species can have very different hearing abilities, a situation that can lead to problems if not taken into consideration. One such case is a series of studies that concluded that laboratory rats were superior to cebus monkeys in discriminating melodies (D'Amato, 1988). However, inspection of the auditory stimuli reveals that some of the melodies contained frequencies below 500 Hz, which, although clearly audible to monkeys, are beyond the hearing range of rats. As a result, the monkeys had to discriminate between two clearly audible, but different

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tunes, whereas the rats had to discriminate between a tune they could hear well and one that contained many inaudible notes. Thus, the results of this study are more likely a demonstration of sensory, rather than cognitive, differences between monkeys and rats. Clearly, it is important not to assume that what is audible to one species is equally audible to another.

Acknowledging species differences is also important for the meaningful interpretation of physiological results, especially as it is not uncommon for physiological data from animals to be compared with behavioral data from humans. In such cases, it appears that there is an underlying assumption that auditory neurons in a particular nucleus have the same general properties regardless of the species in which they are found – indeed, a study may even fail to state what species was being studied (e.g., Skottun, 1998). This situation leads to the construction of auditory models that are composites of different creatures – auditory chimeras – although, because many are half human and half beast, the term "auditory sphinx" might be more appropriate. Such a composite can have interesting results. For example, it has been claimed that single auditory thalamic neurons are able to "distinguish" interaural time differences with the same acuity as human observers (about 10 μs), suggesting that a handful of neurons may account for human localization acuity (Skottun, 1998). However, it turns out that the auditory neurons in question were located in the thalamus of the domestic rabbit, an animal with extremely poor sound localization acuity (~22° vs the ~1° acuity of humans). Thus, although physiologists can detect neural changes to binaural time differences in the rabbit as small as 10 μs, the rabbit can at best detect differences of about 30 μs, assuming that it is relying solely on the time differences available to it at its threshold with no help from spectral cues, which is by no means certain. In short, physiological results must be compared with behavioral results obtained in the same species for the conclusions to be meaningful.

Electrophysiological measures of hearing Because behavioral tests of hearing are difficult and time-consuming, measures of neural responses are often used to estimate what an animal can hear. Some common physiological measures are the cochlear microphonic, thresholds of units in the inferior colliculus, and the auditory brainstem response (ABR). Although such measures cannot help but reflect some aspect of hearing, they are imperfect estimates of actual hearing ability. Moreover, the degree to which they correspond to behavioral thresholds is generally unknown, as few studies have attempted to determine the correspondence between behavioral and electrophysiological thresholds (for an exception, see Szymanski et al., 1999).

An example of the difference between behavioral and electrophysiological results can be seen in the comparison of the behavioral audiogram and ABR thresholds of C57BL/6J × C3HeB/FeJ mice (Koay, Harrington, Heffner, & Heffner, unpublished observations). As shown in Figure 19.5, the ABR generally reflects the behavioral audiogram, even to the extent of indicating the animals' best frequency. However, the ABR thresholds overestimate low-frequency hearing while underestimating high-frequency hearing and best sensitivity. Such a discrepancy is not surprising and there are at least two reasons why one would expect the two estimates of hearing to diverge (Szymanski et al., 1999). First, the tones used to generate the ABR are not pure tones because they

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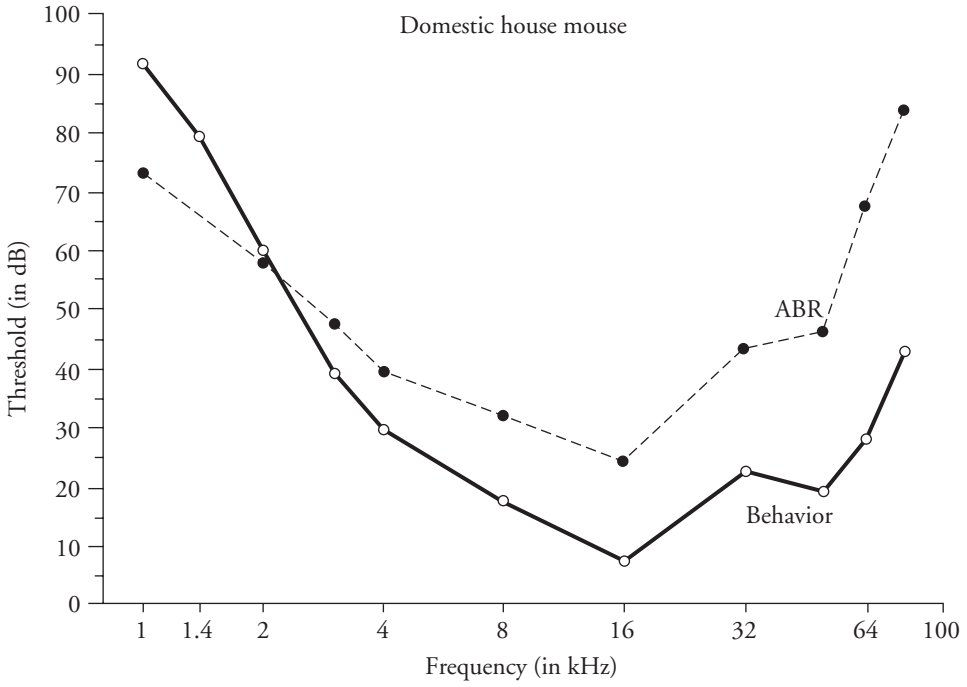


Figure 19.5 Behavioral audiogram and auditory brainstem response (ABR) thresholds of C57BL/6J × C3HeB/FeJ mice. Note that the ABR overestimates low-frequency sensitivity and underestimates high-frequency sensitivity.

have rapid onsets that cause "spectral splatter," whereas the tones used in behavioral audiograms have slow onsets to keep the signal pure. Second, the ABR procedure uses very brief tones with effective durations of 2 ms, whereas behavioral audiograms use much longer tone durations of 400 ms or more. However, detection thresholds depend on the duration of a sound, with sensitivity (in humans) improving as the duration of the stimulus increases up to about 200 ms, a process known as temporal integration (e.g., Moore, 1997). Thus, the difference in the purity and duration of the stimuli alone are sufficient to suggest that the ABR will be an imperfect estimate of behavioral sensitivity.

Role of the Auditory Cortex in Hearing

The search for the functions of different areas of cortex began during the late part of the nineteenth century following the discovery of the motor cortex in the dog by Fritsch and Hitzig (1870). One of the principal localizationists involved in this search was David Ferrier, a British physician who studied the cortex of various mammals using electrical stimulation and ablation techniques. By observing that stimulation of the

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temporal lobe of monkeys resulted in an acoustic startle reaction, Ferrier correctly placed the auditory cortex in the upper two thirds of the superior temporal gyrus (Ferrier & Yeo, 1885). His subsequent demonstration that removal of this area abolished all responsiveness to sound was then taken as conclusive evidence that he had successfully identified the auditory cortex. However, ablation studies conducted by other researchers failed to replicate his findings, with the result that William James concluded that Ferrier was most likely wrong (James, 1890). But, as we shall see, Ferrier was essentially correct in both the location of the auditory cortex and the effect of its ablation (H. E. Heffner, 1987).

Although it is currently fashionable to infer the function of an auditory center from the response properties of its neurons, historically it has been the results of ablation/behavior studies that have carried the most weight (Neff et al., 1975). Although experimental lesions have been made in all auditory centers, only auditory cortex has been studied extensively. The following sections review the effect of auditory cortex lesions on the detection and localization of sound as well as on the discrimination of frequency, including tone patterns and frequency sweeps. The results are presented by species, as the role of the auditory cortex can vary greatly from one species to the next.

Making and assessing the effects of cortical lesions

Making cortical lesions There are two basic methods for making cortical lesions. To date, virtually all experimental lesions of the auditory cortex have been permanent lesions made by subpial aspiration. However, it is also possible to make reversible lesions, that is, to temporarily inactivate the cortex either by cooling it or by the application of chemicals (Lomber, 1999), a technique that is bound to yield new insights into the role of auditory cortex.

Assessing the effects of cortical lesions The effects of cortical lesions have been assessed with various behavioral techniques, including the conditioned suppression technique in which an animal is trained to associate a sound with a mild shock. However, studies on the neural mechanisms of fear conditioning have suggested that disruption of auditory input to the amygdala could reduce or eliminate the response of an animal to a sound that has been paired with shock (e.g., LeDoux, Sakaguchi, & Reis, 1984). Thus the question arises as to whether any of the cortical deficits about to be described could be attributed to reduced fear conditioning rather than to a strictly auditory deficit. The answer to the question is no. Most ablation/behavior studies include control tests to demonstrate that an observed deficit is not due to any attention, motivation, cognitive, or motor disorder. For example, when an animal is unable to discriminate sounds, a routine control test is to use the same procedure to demonstrate that it retains the ability to detect the sounds. Moreover, the loss of the ability to localize sound following cortical lesions can be demonstrated with a pure reward procedure as well as with a shock procedure (e.g., H. E. Heffner & Heffner, 1990b). Thus a reduction in fear conditioning cannot serve as an alternative explanation of the effects of cortical lesions on hearing.

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On the other hand, the possibility that the reduced fear response observed following lesions of the amygdala (and other sites) may be due to a hearing loss has never been ruled out. Not only has the possibility that such lesions themselves may cause a hearing loss never been investigated, but the lesions are made stereotaxically and the earbars used to position an animal's head in a stereotaxic device are known to rupture an animal's eardrums. Thus, although a reduction in fear conditioning cannot account for the cortical deficits in hearing, it is not possible to rule out hearing loss as an alternative explanation of the results of fear-conditioning studies, especially those obtained using stereotaxically placed lesions.

Detection of sound

Ablation of the auditory cortex has a dramatic effect on absolute sensitivity in monkeys and humans, a small effect in carnivores, and little or no effect in other species that have been tested.

Primates The effect of both unilateral and bilateral auditory cortex ablation on absolute thresholds has been studied in some detail in macaque monkeys (H. E. Heffner & Heffner, 1986, 1989, 1990a). Unilateral ablation of the auditory cortex results in a substantial hearing loss in the ear opposite the lesion (the contralateral ear) with no effect on thresholds in the ear on the same side as the lesion (the ipsilateral ear). The hearing loss is greatest immediately after ablation with pure-tone thresholds improving over a period of a few months to near normal levels. The residual hearing loss is small and best demonstrated by comparing preoperative and postoperative thresholds.

Unilateral damage to the auditory cortex in humans undoubtedly results in a contralateral hearing loss (e.g., Karp, Belmont, & Birch, 1969). However, because the residual hearing loss is small, especially when part of the auditory cortex is spared, and because premorbid audiograms of brain damaged patients are rarely available for comparison, the existence of such a hearing loss has been controversial.

Bilateral ablation of the auditory cortex in Japanese macaques results in a profound hearing loss. Initially, there may be total deafness lasting for a few days to a few months after surgery. Pure-tone thresholds show substantial recovery during the first two months, with more gradual recovery thereafter, but the animals continue to show a substantial hearing loss several years after surgery. Thus Ferrier was correct – lesions of the posterior two-thirds of the superior temporal gyrus in monkeys do result in a profound hearing loss. The failure of his contemporaries to replicate his results was most likely because their lesions did not extend far enough into the depths of the Sylvian fissure and were therefore incomplete (H. E. Heffner, 1987).

Although bilateral damage to the auditory cortex in humans is rare, there have been cases showing a hearing loss similar to that observed in macaques. That is, the patient reports a sudden inability to hear any sound followed by gradual, but incomplete, recovery (Jerger, Weikers, Sharbrough, & Jerger, 1969; for a review, see H. E. Heffner & Heffner, 1986).

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This pattern of the cortical hearing loss – deafness followed by substantial but incomplete recovery – suggests that the detection of sound is primarily mediated by subcortical structures that receive input from the auditory cortex. The removal of the cortex, then, results in deafness due to the shock of the sudden loss of cortical input (i.e., diaschisis). This condition is followed by partial recovery as the subcortical areas adjust to the loss of the cortex. There is currently no evidence that the recovery of hearing is mediated by other cortical areas.

Nonprimates The possibility of a cortical hearing loss has been examined in a few other species, none of which have shown the dramatic hearing loss that occurs in primates. Bilateral ablation of all neocortex in the Virginia opossum has no noticeable effect on absolute thresholds, whereas bilateral auditory cortex lesions in rats, cats, dogs, and ferrets result in small, but detectable hearing losses (for a review, see H. E. Heffner & Heffner, 1986).

On the other hand, a recent study found that rats were initially unresponsive to sound following temporary inactivation of auditory cortex using muscimol, a GABA-A agonist with an inhibitory effect on neurons (Talwar, Musial, & Gerstein, 2001). However, no control tests were conducted to determine whether the effect was due to a general unresponsiveness to sensory stimuli, and no conclusion can be drawn as to whether such inactivation of the auditory cortex in rats causes a hearing loss.

Localization of sound

The discovery that auditory cortex ablation in cats results in a sound localization deficit was made over half a century ago by W. D. Neff and his colleagues (Neff, Fisher, Diamond, & Yela, 1956; Neff & Yela, 1948). As with the cortical hearing loss, this deficit is found in some species (primates and carnivores), but not others (rodents).

Primates Unilateral lesions of the auditory cortex result in a complete inability to discriminate the locus of a sound in the hemifield opposite the lesion (the contralesional hemifield), an effect that has been demonstrated in both macaques and squirrel monkeys (H. E. Heffner, 1997; Thompson & Cortez, 1983). The animals retain both the normal ability to localize sound in the ipsilesional hemifield and the ability to distinguish sounds arising in the one hemifield from sounds arising in the other. There is also some residual ability to localize sound in the contralesional hemifield when the source is close to midline (e.g., within 15° of midline), which may be mediated by the intact hemisphere. In short, unilateral ablation results in a "collapse" of auditory space in the contralesional hemifield.

Bilateral ablation of the auditory cortex appears to be the sum of two unilateral lesions – auditory space collapses in both hemifields (H. E. Heffner, 1997; H. E. Heffner & Heffner, 1990b). Although a macaque with a bilateral lesion can be trained to distinguish left *sounds* from right *sounds*, it shows no awareness of the location of the sound *source*. This result is demonstrated by the great difficulty the animals have in learning to approach the source of a continuous sound, which they eventually do by using a tracking

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strategy (i.e., move to where the sound is loudest). Some operated animals eventually learn to go left or right when the sound is brief, but they appear to be learning to associate a spatial response with an arbitrary sound that has no spatial locus. However, the animals are completely unable to distinguish sounds coming from two locations within the same hemifield. Thus, bilateral auditory cortex lesions appear to result in the total collapse of the left and right auditory hemifields leaving only a residual ability to discriminate between left and right sounds that are devoid of spatial meaning.

The effect of bilateral lesions has also been studied in the bushbaby (*Galago senegalensis*), a prosimian (Ravizza & Diamond, 1974). Although the results suggested that their sound localization ability is only moderately affected by auditory cortex lesions, only preliminary results were presented, and it is not known whether the remaining ability represents a species difference or if the lesions were incomplete.

Most studies indicate that unilateral lesions in humans result in a sound-localization deficit in the hemifield contralateral to the lesion, just as they do in macaques (for reviews, see H. E. Heffner & Heffner, 1990b; Neff et al., 1975). However, the observed deficits do not appear to be as severe as those found in macaques, even for patients in which one hemisphere has been entirely removed (Lessard, Lepore, Poirier, Villemagne, & Lassonde, 2000). One possible explanation is that such patients typically sustained cortical damage early in life, allowing for greater function compensation than occurred in the macaques, all of which were adults at the time of surgery. Surprisingly, a recent study has suggested that sound localization in humans is lateralized such that lesions of right, but not left, auditory cortex result in a sound localization deficit (Zatorre & Penhune, 2001). However, these results rest on the premise that partial lesions of the auditory cortex that include primary auditory cortex, AI, are sufficient to produce the classic sound localization deficit – but this premise may not be correct. In macaques, partial lesions of the auditory cortex, even if they include substantial portions of primary auditory cortex, do not result in a total collapse of the contralateral auditory field (Harrington & Heffner, 2002). Thus the results of the study by Zatorre and Penhune (2001) should be reevaluated with regard to the location and completeness of the lesions.

The effects of bilateral lesions in humans on sound localization are not clear owing to the rarity of cases with complete bilateral auditory cortex lesions. Indeed, given the severe hearing loss that accompanies such lesions, the patients would be difficult to examine. However, we expect that bilateral lesions in humans, as in macaques, would result in a complete inability to localize sound.

Carnivores The effect of cortical lesions on sound localization has been studied in cats, dogs, and ferrets (see H. E. Heffner & Heffner, 1990b for a review). All species appear to show the same sound-localization deficit as macaques. Specifically, unilateral lesions in cats and ferrets result in a sound-localization deficit in the hemifield contralateral to the lesion, and bilateral lesions in cats, dogs, and ferrets result in a deficit in both hemifields. In addition, a study using cats indicated that restricting the lesion to a particular frequency representation in primary auditory cortex affects the ability to localize those specific frequencies (Jenkins & Merzenich, 1984). However, this finding deserves further study in light of the finding that cats experience a mild hearing loss

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following cortical lesions and because lesions restricted to primary auditory cortex in monkeys do not appear to result in the classic sound-localization deficit (see above).

Rodents In contrast to primates and carnivores, bilateral ablation of the auditory cortex in the Norway rat, as well as in the wild wood rat (*Neotoma floridana*), does not abolish the ability to localize sound – at most it may result in a slight increase in threshold (e.g., H. E. Heffner & Heffner, 1990b; Kavanagh & Kelly, 1986).

Other species Two other species that have been examined are the hedgehog, an insectivore, and the Virginia opossum, a marsupial (Ravizza & Diamond, 1974; Ravizza & Masterton, 1972). In both cases, cortical ablation appears to result in increased thresholds for left–right discriminations. However, neither study examined the possibility that, in spite of the animals' ability to perform a left–right discrimination, the lesions may have resulted in a collapse of auditory space within each hemifield. Thus the complete effect of cortical lesions on sound localization in these species is not known.

The discrimination of frequency

Early electrophysiological studies indicated that frequency was mapped on the auditory cortex in an orderly manner (tonotopic maps), giving rise to the idea that the cortex is necessary for frequency discrimination (Neff et al., 1975). Once it was discovered that cortical ablation did not totally abolish the ability of the animals to discriminate one frequency from another, testing moved on to the discrimination of tone patterns and frequency sweeps.

Discrimination of discrete frequencies Studies of the effect of auditory cortex lesions on the ability of macaques to discriminate tones of different frequency have established that auditory cortex lesions result in a small but reliable increase in thresholds (e.g., Massopust, Wolin, & Frost, 1970). For example, the average frequency increment needed to discriminate a 625 Hz tone from higher frequencies was shown to increase from 7.5 Hz to 27.5 Hz (Harrington, Heffner, & Heffner, 2001). Thus, although the auditory cortex is not necessary for frequency discrimination per se, its loss does result in an increase in thresholds.

Although an initial study indicated that bilateral auditory cortex lesions in cats abolished the ability to discriminate frequency (Meyer & Woolsey, 1952), a subsequent study failed to find a deficit (Butler, Diamond, & Neff, 1957). However, the two studies used different methods of stimulus presentation, giving rise to the idea that although operated animals could detect a *change* in the frequency of an ongoing train of tone pips, they could not discriminate tones of two different frequencies if the presentation of the tones was separated by a long silent interval – that is, they could make a "relative," but not an "absolute," frequency discrimination (Thompson, 1960). However, it has since been demonstrated that cats with auditory cortex lesions retain the ability to make absolute judgments of frequency (Cranford, 1978) and the current view is that such lesions have at most only a small effect on the ability of cats to discriminate frequency.

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Although it was established long ago that removal of cortex has no effect on frequency discrimination in the Norway rat (J. B. Kelly, unpublished doctoral dissertation), a recent study has suggested that the chemical inactivation by the application of muscimol to the auditory cortex results in a temporary inability to discriminate frequency (Talwar et al., 2001). The implication of this finding is that the auditory cortex in rats (and other mammals) is normally involved in the discrimination of frequency. However, no control tests were conducted to rule out alternative explanations of the failure of the rats to respond to a difference in frequency. Thus we do not know at this time if the results were due to an auditory deficit per se, or to an attentional, cognitive, or motor deficit. Indeed, it is conceivable that the application of muscimol to the auditory cortex results in the perception of phantom sound that distracts the animal from external auditory stimuli.

Tone pattern discrimination Once it was found that cortical lesions did not abolish the ability to discriminate frequency, researchers moved on to determine the role of the cortex in discriminating changes in temporal patterns of tones. The tone patterns were typically sequences of three tone pips of a high or low frequency, such as Low–High–Low, High–Low–High, Low–Low–Low, and High–High–High. By using such stimuli, researchers hoped to discover how the cortex processed patterns of stimuli that were analogous to the spatial patterns used in visual and somatosensory studies. However, because many of the discriminations proved difficult even for normal animals to learn, the resulting deficits may have been cognitive rather than perceptual in nature. Thus the significance of this line of research is unclear (for reviews, see Elliott & Trahiotis, 1972; Neff et al., 1975).

Frequency sweeps The discovery that some auditory cortex neurons in cats are selective for the direction of a frequency change (Whitfield & Evans, 1965) was the motivation for investigating the effect of auditory cortex lesions on the discrimination of frequency sweeps. The results of these studies indicated that bilateral auditory cortex lesions impaired, but did not abolish, the ability to discriminate a rising from a falling frequency sweep (Kelly & Whitfield, 1971).

The motivation for determining the effect of cortical lesions on the discrimination of frequency sweeps by macaques was different. Specifically, Japanese macaques lose the ability to discriminate between different forms of their coo call following auditory cortex lesions (H. E. Heffner & Heffner, 1986, 1994, 1995b). Although this result suggested that the animals had an aphasia-like deficit, in that they were no longer able to interpret their vocal communications, the possibility existed that it might be part of a general sensory deficit. Because the coos used in those studies were tonal calls that either rose or fell in frequency, it was necessary to determine whether auditory cortex lesions affected the ability to discriminate frequency sweeps. The results indicated that although the animals could discriminate a rising from a falling frequency sweep, they did so on the basis of absolute frequency differences, for example, comparing the initial frequency of each sweep, rather than responding to the direction of frequency change. When they were prevented from using that strategy, by randomizing the frequency of the stimuli, their performance fell to chance. Thus, auditory cortex ablation in macaques abolishes

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the ability to determine whether a sound is *changing* in frequency (Harrington et al., 2001). This result demonstrates that the inability of macaques to discriminate their coo vocalizations is part of a broader sensory deficit. It also raises the question of whether a similar deficit underlies sensory aphasia in humans.

Recent studies have found that cortical lesions impair the performance of gerbils in discriminating rising from falling frequency sweeps (Ohl, Wetzel, Wagner, Rech, & Scheich, 1999; Wetzel, Ohl, Wagner, & Scheich, 1998). However, the group data presented in these studies (no individual data were shown) showed relatively large variance, suggesting that not all of the operated animals were impaired. It has long been the rule that a lesion must consistently result in a deficit before the ablated area can be considered essential for a function (James, 1890). Thus, without knowledge of individual results, no definite conclusion can be reached regarding the role of auditory cortex in gerbils on the discrimination of frequency sweeps.

Final considerations regarding the auditory cortex

The auditory cortex has been described as being functionally unilateral, with each hemisphere processing sound arising from the contralateral hemifield (Glendenning & Masterton, 1983). This view is based on the observation that the majority of neurons in the auditory cortex respond best to sounds in the contralateral sound field, that ipsilateral input is usually inhibitory, and that unilateral damage to the auditory cortex results in sound-localization deficits confined to the hemifield contralateral to the lesion. Although this view may be valid for sound localization, it does not necessarily apply to other auditory abilities, such as sound detection and discrimination. For these abilities, the auditory cortex may be more appropriately characterized as being asymmetrically bilateral with each hemisphere having a greater involvement in processing information from the contralateral *ear* as opposed to the contralateral hemifield.

With regard to the detection of sound, a unilateral auditory cortex lesion in a macaque results in a hearing loss in the ear contralateral to the lesion, with thresholds in the other ear completely unaffected. This means that sounds arising from the contralesional hemifield, even if inaudible in the ear on that side because of the unilateral hearing loss, will be detected as long as the sound can reach the other ear (H. E. Heffner & Heffner, 1989). The same applies to the discrimination of frequency change, which is impaired for sounds presented to the contralesional *ear* and not for sounds presented in the contralesional *hemifield* that reaches both ears (H. E. Heffner and Heffner, 1994). In neither case, however, is the deficit as severe as that resulting from a bilateral lesion, demonstrating that each hemisphere plays a role in processing sounds from both ears with the opposite hemisphere having a greater role. Thus, although unilateral lesions abolish sound localization in the contralateral hemifield, they affect the detection and discrimination of sounds in the contralateral ear.

It should be noted that the above results are species-specific – they apply to macaques, but not necessarily to other mammals, and particularly not to rodents. There is currently no explanation for why some species should show a deficit when others do not. It may be noted that rats have smooth (lissencephalic) brains, whereas carnivores and primates

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have highly convoluted (gyrencephalic) brains, and that rats normally have poorer sound localization acuity than primates and carnivores. Whether either factor is related to the observed species differences in cortical function is not known. However, it should be clear that a person cannot speak of the function of "the auditory cortex" without also stating the species from which the functions have been inferred.

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